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Universal Soft Coulomb Gap Governs Thermoelectric Performance in Doped Conjugated Polymers

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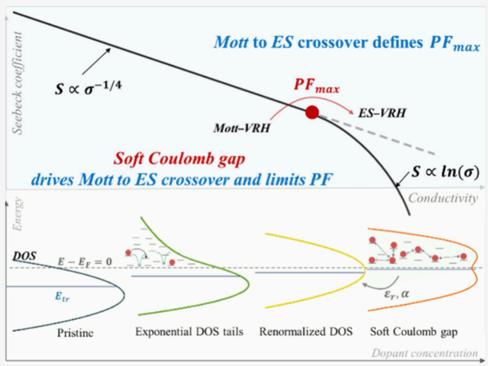
ABSTRACT: **The power factor (PF) in doped conjugated polymers is fundamentally constrained by a trade-off between electrical conductivity and the Seebeck coefficient, yet the underlying universal limit remains unclear. Here, we identify a soft Coulomb gap near the Fermi level as the intrinsic origin of this limit, arising from long-range Coulomb repulsion among localized carriers. Combining systematic experiments and kinetic Monte Carlo simulations across chemically diverse materials, we consistently show that the maximum PF coincides with a distinct transition from Mott to Efros−Shklovskii variable-range hopping, marking the onset of gap formation accompanied by Seebeck coefficient suppression. By increasing the dielectric constant or promoting charge delocalization, the soft gap can be effectively suppressed, shifting the optimal doping range and enhancing performance. These findings establish a unifying physical framework and actionable design rules for overcoming the PF limit, with broad implications for the design of high-performance thermoelectric and other doped organic electronic materials**



Conjugated polymers (CPs) have garnered increasing attention as promising organic thermoelectric (OTE) materials due to their solution processability, mechanical flexibility and intrinsically low thermal conductivity.[1−6] Commonly, the performance of thermoelectric materials is evaluated using the dimensionless Figure of merit ZT, given by $ZT = \frac{\sigma S^2}{\kappa}T$, where σ is the electrical conductivity, S is the Seebeck coefficient, κ is the thermal conductivity, and T is the temperature. Over the past decade, research on organic OTEs has advanced ZT to values around unity at room temperature,[7] a level nearly equivalent to that of their inorganic counterparts. However, as Brunetti et al. pointed out,[8] the field of OTEs appears to be stuck in terms of power factor ($PF = \sigma S^2$), and new approaches for rational material design are urgently needed. This bottleneck can be attributed to a limited comprehensive understanding of the complex interplay of numerous interrelated factors. Among these, the relationship between the doping level, charge transport mechanisms, and the resulting thermoelectric performance is particularly critical, yet poorly understood.

Owing to the disordered nature and ultralow charge concentration of intrinsic CPs, a common and effective strategy to enhance their electrical conductivity is molecular doping, which elevates the carrier density but often suppresses S.[9−12] Hence, striking a balance between σ and S is essential for achieving high-efficiency OTEs. When doped over a sufficient range, a transition regime typically occurs in the S - σ correlation.[13,14] At low doping concentrations until close to the maximum power factor, PF_{max}, the entire thermopower versus conductivity curve can be described by the well-known empirical power law $S \propto \sigma^{-0.25}$ as identified by Glaudell et al.,[15] although deviations have been observed, e.g. for low doping systems.[16] The latter can be attributed to transport by VRH in a density of states (DOS) that is modified by Coulomb traps formed by ionized dopants.[17] In the intermediate doping regime, Li et al. indicated that the DOS is renormalized due to the overlapping Coulomb potentials of homogeneously distributed ionized dopants.[18] In consequence, transport at these dopant concentrations can, in stark contrast

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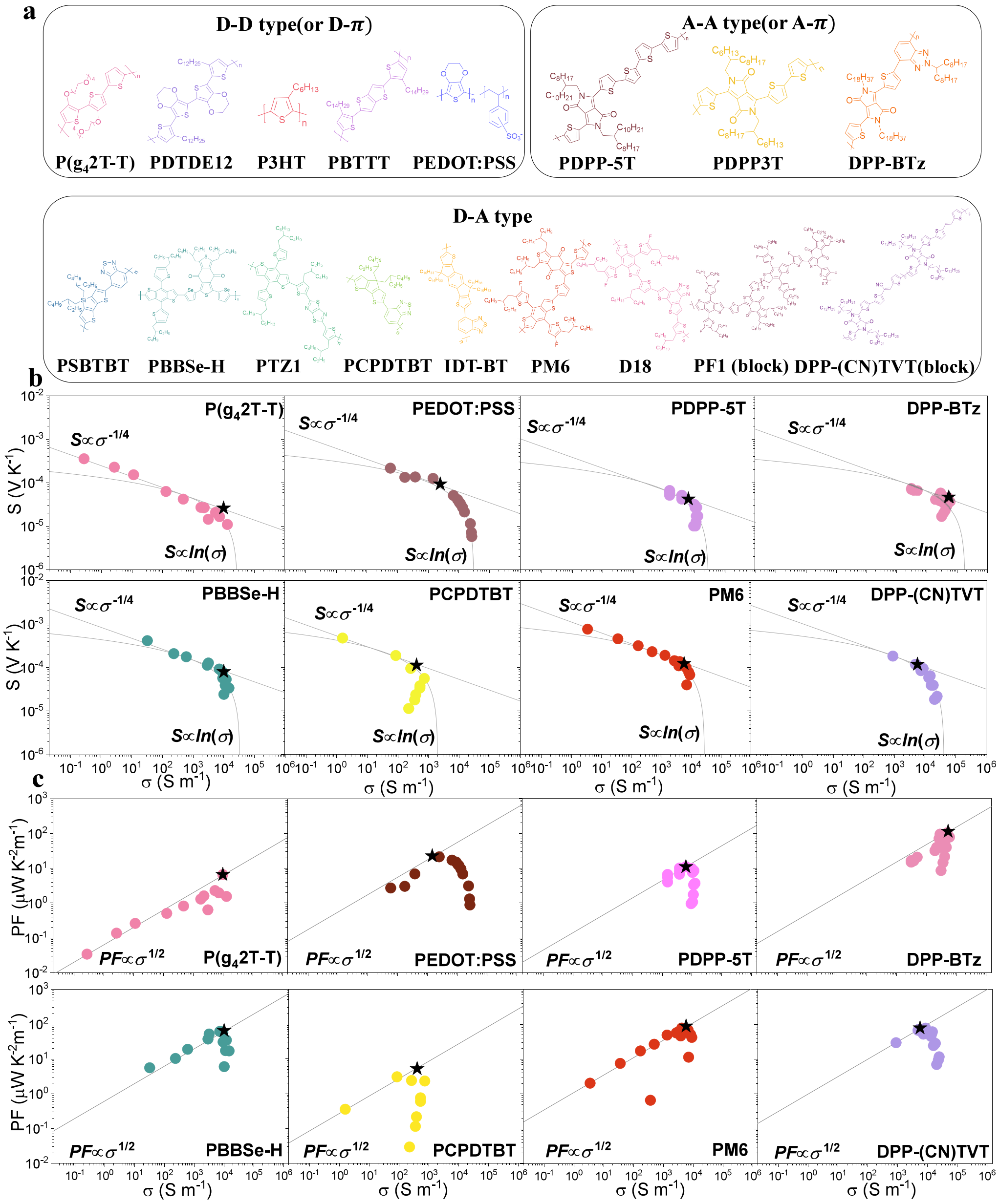

Figure 1. (a) Chemical structure of the studied materials in this work, representing diverse backbone designs including donor−donor, donor−acceptor, acceptor−acceptor, and copolymer architectures. (b) Experimental Seebeck coefficient-conductivity (*S*-*σ*) curves and (c) corresponding power factor (PF) as a function of conductivity for representative polymers under p-type doping conditions. P($g_4$2T-T),[24] PEDOT:PSS,[28] PDPP-5T,[29] DPP-BTz,[31] PBBSe-H, PCPDTBT, PM6, and DPP-(CV)TVT.[33]

to low-doping concentrations, be described as transport of free charge carriers, not being limited by charge trapping by the Coulomb potentials of the dopant counterions. Importantly, the maximum power factor is typically achieved in this regime.[14,19] Beyond PF_{max}, the *S*-*σ* correlation instead follows $S \propto \ln(\sigma)$, as also observed in experiments.[12,15,20] Recently, Yang et al. indicated the Seebeck coefficient follows Heike's formula at intermediate to high doping levels and attributed this to the conductivity becoming limited by carrier−carrier repulsion.[12] However, the underlying physics remained unclear, and a comprehensive understanding of how the transition affects the optimal PF is lacking.

Here, we combine experimental measurements and kinetic Monte Carlo simulations to elucidate the fundamental mechanisms governing the Seebeck coefficient−conductivity (*S*-*σ*) correlation in doped conjugated polymers across the entire doping range. We demonstrate that the transition from the empirical power law $S \propto \sigma^{-0.25}$ to the logarithmic dependence $S \propto \ln(\sigma)$ universally originates from the formation of a soft Coulomb gap in the renormalized Gaussian density of states, reflecting the dominance of carrier−carrier Coulomb interactions.[12,21] Although this transition is universally observed, its precise manifestation depends critically on the material's dielectric properties and energetic disorder. Shifting the *S*-*σ* roll-off to higher doping levels is identified as a key strategy for enhancing the maximum power factor. Accordingly, we propose that enhancing dielectric screening by increasing the relative permittivity and promoting charge delocalization to mitigate Coulomb interactions represent effective design strategies for high-performance organic

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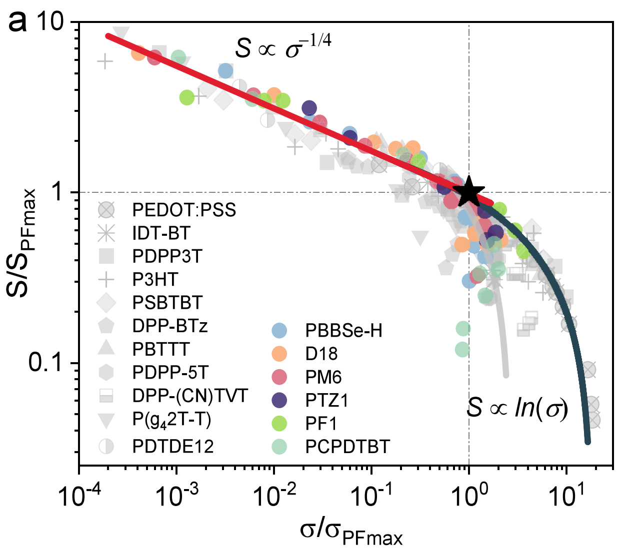

Figure 2. (a) Normalized Seebeck coefficient (S/S_{max}) and (b) normalized power factor (PF/PF_{max}) as a function of conductivity ((σ/σ_{max})) for all studied polymers. Both the Seebeck coefficient (S) and conductivity (σ) are normalized to their respective values at maximum power factor. The star symbol represents the maximum PF (PF_{max}). All data are derived from the measurements presented in Figure 1. Solid lines are trend lines following the indicated functional form.

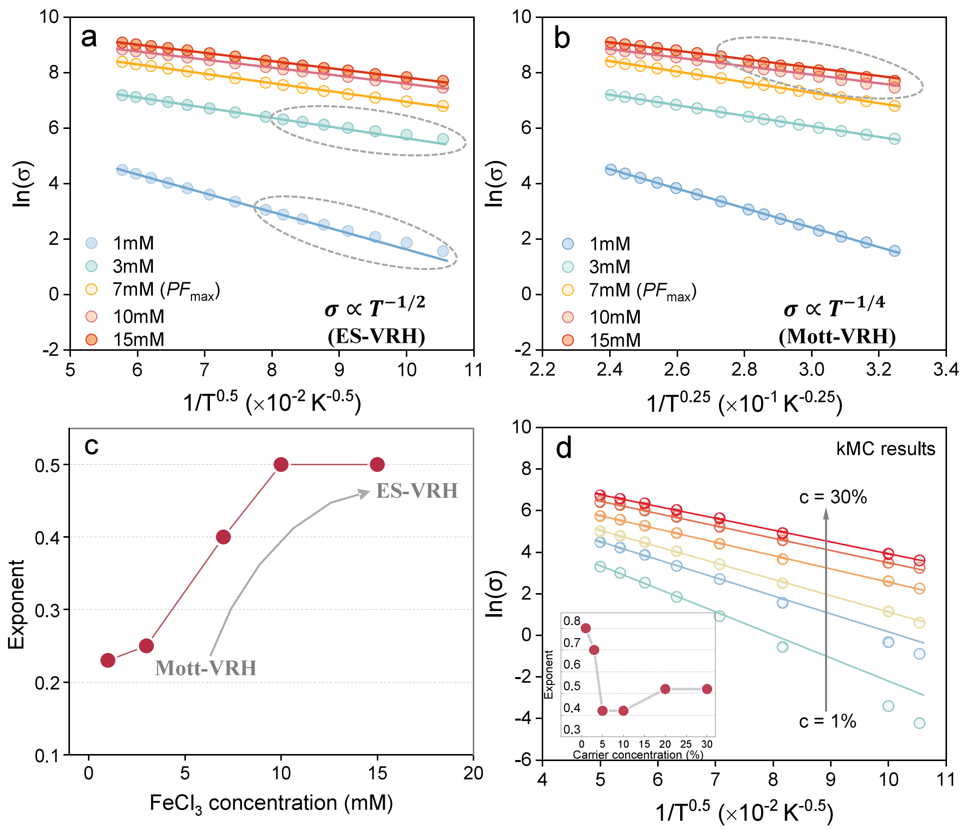

Figure 3. Temperature-dependent conductivity in experiments and simulations. (a-b) Experimental plots of logarithmic conductivity as a function of temperature for material PBBSe-H at different $FeCl_3$ concentration, showing characteristic temperature exponent 0.5 and 0.25, respectively. (c) Extracted optimal hopping exponent α as a function of $FeCl_3$ concentration, indicating a clear transition from Mott-VRH ($\alpha \sim 0.25$) to ES-VRH ($\alpha \sim 0.5$) transport with increasing doping level. (d) Kinetic Monte Carlo (kMC) calculations of logarithmic conductivity as a function of temperature at different hole concentration. Default kMC parameters: intersite distance a_{NN} = 1.8 nm, energetic disorder σ_{DOS} = 75 meV, attempt-to-hop frequency $\nu_0 = 1 \times 10^{13}$ s^{-1}, relative dielectric constant ε_r = 3.6, temperature T from 100 to 400 K. Lines are fits to ln $\sigma \propto T^{-0.5}$. Circles in panels (a) and (b) highlight nonlinear fits corresponding to Mott-VRH and ES-VRH regimes at different doping levels.

thermoelectric materials. These findings establish a robust mechanistic framework for rationally tailoring electronic structure and charge transport to optimize thermoelectric performance.

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The materials investigated in this study are depicted in Figure 1a. To enable a comprehensive and systematic exploration of the structure−property relationship governing thermoelectric performance, we selected a series of conjugated polymers (CPs) featuring diverse conjugated backbones. These CPs incorporate various molecular design motifs, including donor−donor (D−D), donor−acceptor (D−A), acceptor−acceptor (A−A) motifs, and block copolymer structures, thereby offering a broad platform for correlating molecular structure with charge transport and thermoelectric behavior. The thermoelectric properties for all materials were evaluated by measuring the conductivity and Seebeck coefficient on lateral (in-plane) devices, following procedures outlined in detail in our previous works[19,22,23] or adapted from relevant literature sources.[24−33] $FeCl_3$, with a LUMO below −5.8 eV,[34,35] was employed as the (p-type) dopant for all polymers in this study. More details are provided in the Experimental Section. The resulting thermoelectric properties of representative materials for each polymer type are presented in Figure 1b (S−σ) and Figure 1c (PF−σ). The complete data set for all materials is shown in Figure S3 and summarized in Table S2 of the Supporting Information. All these data highlight a general trend in the Seebeck coefficient−conductivity (S−σ) relationship, where the power factor reaches a maximum, followed by a nonlinear decrease in the Seebeck coefficient at higher conductivities.

To better understand the relationship between chemical structure and thermoelectric performance and its influence on charge transport, we introduce a new representation of thermoelectric parameters, in which both S and σ are scaled relative to their respective values at maximum power factor, denoted as S_{PFmax} and σ_{PFmax}, respectively. The resulting trends are shown in Figure 2. It reveals that the Seebeck coefficient−conductivity (S-σ) curves universally follow the well-known empirical relationship $S \propto \sigma^{-0.25}$ in the low conductivity, i.e., low-doping, regime. This behavior was initially identified by Glaudell et al.,[15] and the underlying physical mechanism was subsequently elucidated by Abdalla et al.,[17] who attributed it to charge transport being predominantly governed by an exponential tail of deep states in the intrinsically Gaussian DOS due to the attractive Coulomb potential surrounding ionized dopants. However, beyond a certain point, this relationship transitions to a logarithmic dependence, $S \propto \ln(\sigma)$, indicating a shift in the dominant charge transport mechanism. In addition, the $S \propto \ln(\sigma)$ trend exhibits a strong material dependence, and as such is not universal.

To elucidate the origin of the transition regime in the S-σ correlation, we conducted temperature-dependent conductivity measurement on lateral (in-plane) devices for all studied materials across various doping concentrations and temperature gradients. The temperature dependence of the electrical conductivity in a disordered system can generally be described by

$$\sigma = \sigma_0 \exp\left[-\left(\frac{T_0}{T}\right)^{\alpha}\right] \tag{1}$$

where σ_0 denotes the prefactor, and T_0 is the characteristic temperature. Both σ_0 and T_0 are material- and film-dependent parameters influenced by factors such as the electronic density of states, Debye frequency of the material, localization length and dielectric constant.[36] The exponent α determines the (stretching of the) temperature dependence of the conduction and depends on the charge transport mechanism. Specifically, for a more or less constant DOS around the Fermi level, $\alpha = 1/(1 + d)$ with d the dimensionality of the system, for 3D systems leading to Mott-type variable range hopping (Mott-VRH), with $\ln \sigma \propto T^{-0.25}$. For systems that are not evidently 2D, the exponent $\alpha = 0.5$ is typically attributed to the opening of a soft Coulomb gap at the Fermi energy, resulting in Efros-Shklovskii variable-range hopping (ES-VRH).[37,38]

As shown in Figure 3a, the conductivity of the material PBBSe-H does not exhibit a good linear fit in the low doping regime. In contrast, the conductivity versus temperature curve shows an excellent linear trend, that is, $\alpha = 0.5$, starting at a doping concentration of ∼7 mM $FeCl_3$, which corresponds to the maximum PF (PF_{max}) value. This indicates that the charge transport follows the ES-VRH model at and beyond the PF_{max}. Interestingly, prior to PF_{max}, the conductivity versus temperature curve exhibits an excellent linear trend with an exponent of 0.25, as shown in Figure 3b, suggesting conventional Mott-VRH behavior. In that representation, deviations from a linear trend are observed at higher doping concentrations. Hence, for the material PBBSe-H, across the doping regime from 1 mM to 15 mM and the temperature range from 90 to 300 K, a clear change in the exponent for the relation $\ln(\sigma) \propto T^{-n}$ is observed, shifting from 0.25 to 0.5 across the entire doping concentration range, with the transition occurring at the doping concentration of PF_{max}, as shown in Figure 3c and, in more detail, in Figure S11. The change in exponent indicates a transition in the charge transport mechanism from Mott-VRH to ES-VRH. Similar observations of the exponent transitioning from 0.25 to 0.5 at PF_{max} were found in the other materials systems, as depicted in Figures S4−S15. This suggests a rather general transition in the charge transport mechanism from Mott-VRH to ES-VRH for heterogeneous systems, including doped CPs, with the transition consistently occurring around PF_{max}. Simultaneously, the S-σ correlation changes from the well-known empirical power law $S \propto \sigma^{-0.25}$ to $S \propto \ln(\sigma)$, as shown in Figure 2. We note that, in the regime between Mott-VRH and ES-VRH, the two transport mechanisms may coexist or blend. Furthermore, the evolution of the corresponding exponents in this intermediate regime may be influenced by doping-induced variations in the (crystalline) morphology of different polymers.

To further investigate the universality of the observed transition in the charge transport mechanism from Mott-VRH to ES-VRH for doped CPs, kinetic Monte Carlo (kMC) simulations were performed. The employed model has been described in detail in previous studies.[22,39−41] In brief, it gives a numerically exact description of Coulombically interacting localized particles, hopping on a fixed lattice with site energies drawn from a Gaussian DOS. The calculated temperature-dependent conductivity as a function of charge carrier concentration is presented in Figure 3d. These results are consistent with the experimental data, showing that the conductivity versus temperature curve exhibits an excellent linear trend with an exponent of 0.5 at and beyond the concentration of optimal PF (∼10%). This finding further confirms the universal charge transport trend in doped CPs.

The kMC simulations also reproduce the lower exponents (<0.5) at lower doping levels, as shown in Figure 3d (inset) and Figure S16 in the SI. However, the upswing in exponent at the lowest doping levels is not visible in the present experiments. We attribute this to the fact that all simulations are performed for a strictly Gaussian DOS, as is appropriate in

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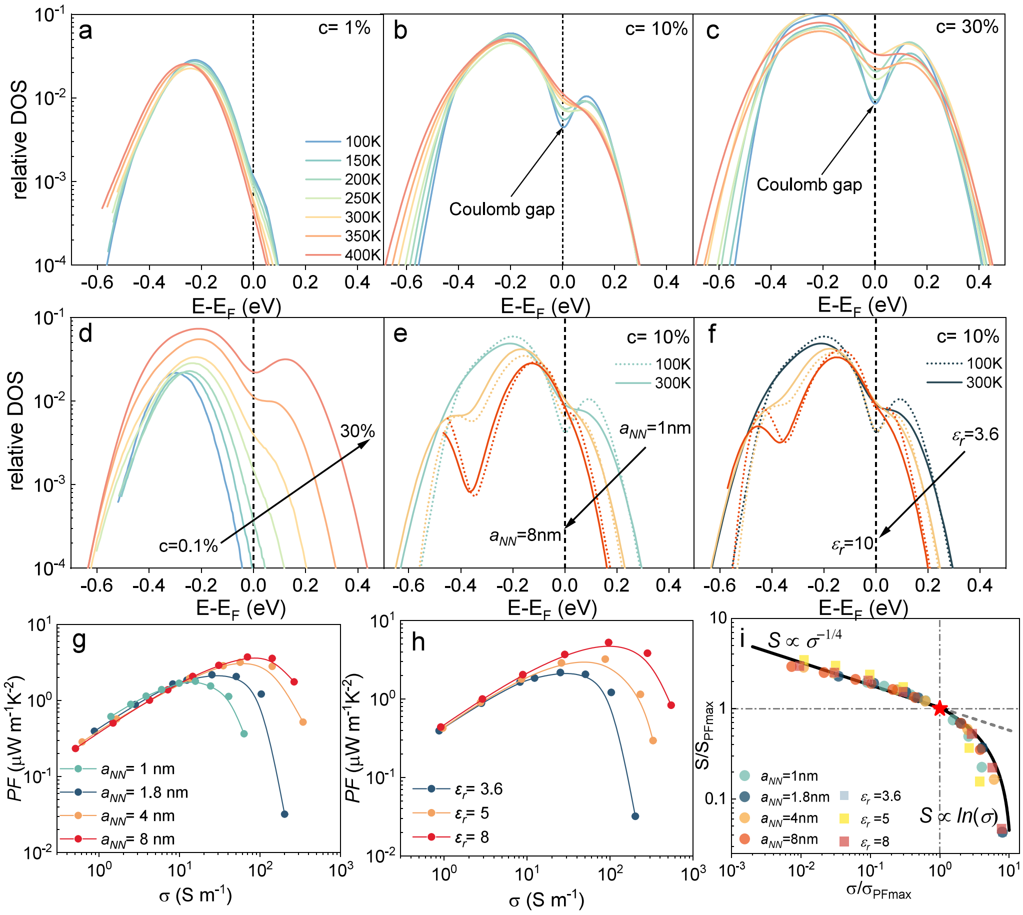

Figure 4. Kinetic Monte Carlo simulations of density of states (DOS) and thermoelectric behavior. (a−c) kMC-calculated evolution of the DOS as a function of temperature at different hole concentrations: (a) 1%, (b) 10%, and (c) 30%. (d) DOS evolution as a function of hole concentration (1−30%) at 300 K. (e) DOS evolution as a function intersite distance (a_{NN}) ranging from 1 to 8 nm, simulated at 100 and 300 K. (f) Evolution of DOS as a function of relative dielectric constant (ε_r) from 3.6 to 8 nm, simulated at 100 and 300 K. (g, h) Thermoelectric characteristics from kMC simulations at varying parameters for (g) a_{NN} = 1, 1.8, 4, and 8 nm; and (h) ε_r = 3.6, 5, and 8. (i) Normalized Seebeck coefficient versus conductivity based on the simulation results in panels (g) and (h). Default kMC simulation parameters (unless otherwise specified): a_{NN} = 1.8 nm, σ_{DOS} = 75 meV, $\nu_0 = 1 \times 10^{13}$ s^{-1}, ε_r = 3.6, and T from 100 to 400 K. For DOS calculations in (a)−(c), the same input parameters as in Figure 3d were used.

the transition regime and beyond, as this is our region of interest.[18] At lower concentrations, the DOS shape is more complicated due to the dopant-induced trap states, as discussed in the introduction.[17] Moreover, the experiments may simply not reach these low doping levels.

Before discussing the possible underlying reasons for the transition in the charge transport mechanism from Mott-VRH to ES-VRH in doped CPs with increasing doping concentration, we will first clarify the differences between Mott-VRH and ES-VRH in disordered organic systems. Both models describe charge transport in strongly disordered systems via hopping between localized states. However, they differ in the physical mechanisms that are accounted for and hence have different conditions for applicability, primarily depending on whether Coulomb interactions play a significant role and how these interactions affect the DOS. Specifically, ES-VRH emerges when strong Coulomb repulsion, arising from carrier−carrier interactions, suppresses the DOS near the Fermi level, leading to the formation of a soft Coulomb gap, which subsequently governs the charge transport behavior. Moreover, temperature has a significant influence on the soft Coulomb gap. As temperature increases, charge carriers gain sufficient thermal energy to overcome the Coulombic suppression near the Fermi level, resulting in a washing-out of the gap and a corresponding restoration of the original DOS. Consequently, the width and depth of the gap decrease with increasing temperature, thereby diminishing its influence on charge transport.

To verify whether indeed the observed change in exponent is due to a transition in transport from Mott-VRH to ES-VRH, in turn originating from the formation of soft Coulomb gap with increasing doping concentration, the detailed evolution of the DOS with temperature was investigated by kMC simulations. As shown in Figure 4a, for a carrier concentration c = 1%, there is negligible change in DOS near the Fermi energy as the temperature increases from 100 to 400 K. However, when c increases to 10% (Figure 4b), a distinct parabolic dip appears in the DOS near E_F around ∼200 K, indicating the formation of a soft Coulomb gap. As the temperature increases to 400 K, this soft Coulomb gap gradually disappears. Combining these observations with the temperature-dependent conductivity behavior shown in Figure 3d, it is evident that at c = 1%, no soft Coulomb gap is formed, and the charge transport is dominated by the Mott-VRH model, characterized by $\ln(\sigma) \propto T^{-0.25}$. As c increases to 10%, the carrier−carrier repulsions become significant, leading to

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the formation of soft Coulomb gap and a corresponding transport behavior of $\ln(\sigma) \propto T^{-0.5}$. With a further increase in c up to 30%, the soft Coulomb gap becomes even more pronounced, as shown in Figure 4c,d.

The observed transition from Mott-VRH to ES-VRH, indicated by the exponent shift from −0.25 to −0.5, appears to be a generic phenomenon across various doped conjugated polymers. Our kMC simulations suggest this universality arises because of the formation of a soft Coulomb gap and thus the onset of ES-VRH primarily depend on a small set of fundamental parameters: specifically, the density of states (DOS) near the Fermi level, determined indirectly by the nearest-neighbor distance (a_{NN}), the width of the Gaussian DOS (σ_{DOS}), and the material's dielectric constant (ε_r). Importantly, these parameters typically exhibit only algebraic variations across different polymer systems, rather than exponential changes. Consequently, a similar DOS renormalization profile, and thus a soft Coulomb gap, is consistently expected to emerge within a doping range of ∼5−20%. To further substantiate this point, we performed additional simulations exploring DOS evolution for different a_{NN}, σ_{DOS}, and ε_r values. The results (shown in Figures S18−S21 of the Supporting Information) confirm that the soft Coulomb gap consistently appears within this doping regime, implying that the ES-VRH transport regime should generally be observable provided sufficient doping levels can be experimentally realized. Indeed, our experimental data aligns well with this prediction, clearly exhibiting a transition from exponent −0.25 to −0.5 whenever doping extends sufficiently beyond PF_{max}.

Notably, as shown in Figure 4e,f, the formation of the soft Coulomb gap is suppressed with increasing a_{NN} (and indirectly the localization length) and ε_r, both of which effectively reduce the Coulomb interaction strength. This trend aligns with the Coulomb interaction equation ($E_C = e/4\pi\varepsilon_r\varepsilon_0 a$), where greater spatial separation between carriers or permittivity reduces the effective Coulomb repulsion, thereby delaying the onset or diminishing the depth of the Coulomb gap. Consequently, the PF is correspondingly enhanced, as shown in Figure 4g,h. Furthermore, as shown in Figure 4i, the normalized S-σ correlation obtained from kMC simulations closely reproduces the experimental trends shown in Figure 2. This agreement further confirms the universality of the observed transition in charge transport mechanism, from Mott-VRH to ES-VRH, governed by the formation of a soft Coulomb gap in doped CPs. We note that as the carrier (or dopant ion) concentration increases, the electrostatic potential between localized charges becomes screened, which has a similar effect to that of increasing the dielectric constant.[42] This enhanced charge screening significantly reduces carrier−carrier repulsion, thereby suppressing the formation of the soft Coulomb gap in the DOS. As shown in Figure 4f, the gap near the Femi level gradually disappears with higher dielectric constant.

Finally, to put our results in perspective, we illustrate the interrelationship between the density of states, charge transport, and thermoelectric properties across the entire doping regime. In general, the detailed characteristics of the DOS determine charge transport in disordered organic systems and thereby play a dominant role in shaping the corresponding thermoelectric properties. Figure 5 illustrates the evolution of the DOS profile and the associated S-σ correlation with increasing doping concentration. As discussed previously in this paper, at low doping concentrations, the overall DOS, featuring an additional exponential tail, introduced by dopant ions, below an otherwise Gaussian DOS, governs charge transport. In this regime, charge carriers follow the conventional VRH mechanism, resulting in the empirical power law of $S \propto \sigma^{-0.25}$. With increasing doping concentration, up to intermediate levels, the overlapping Coulomb potentials of uniformly distributed ionized dopants progressively renormalize the DOS, enabling (effectively) free-carrier transport rather than trap-limited hopping. Notably, the optimal power factor is typically achieved in this regime. Unlike in the previous regime, the Fermi and transport energies lie in a region where the DOS is no longer strongly energy dependent, leading to Mott-like VRH (eq 1 with $\alpha \approx 1/4$). It is noted that the exponent may deviate from 1/4 in the low doping regime (i.e., $c < 10^{-3}$), where charge transport is governed by the intrinsic DOS of the host material. As indicated by Abdalla et al.,[17] this deviation occurs regardless of whether the transport follows the nearest-neighbor hopping (NNH) or VRH model under such low doping conditions. As the doping level increases, the dopant-ion-induced DOS gradually becomes dominant in governing charge transport, and consequently, the characteristic 1/4 exponent emerges. As the doping concentration continues to increase, the carrier−carrier interactions become significantly more pronounced, leading to the formation of a soft Coulomb gap around the Fermi level in the renormalized DOS, which governs the charge transport and induces a transition to the ES-VRH model (eq 1 with $\alpha \approx 1/2$). Consequently, the S-σ correlation shifts to a logarithmic dependence, described by $S \propto \ln(\sigma)$. It is worth noting that the $S \propto \ln(\sigma)$ relationship is strongly material-dependent, i.e., it is not universal and is significantly influenced by the degree of disorder in the material.

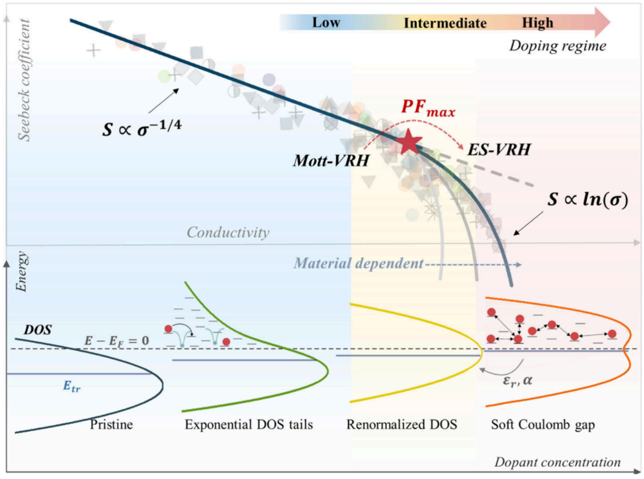

Figure 5. Doping-driven evolution of the density of states (DOS) and its impact on the Seebeck-conductivity (S-σ) correlation. As doping concentration increases, the DOS progressively evolves from an intrinsic Gaussian distribution to the formation of exponential tails, then transitions into a renormalized DOS, and ultimately develops into a Coulomb-gap feature. This DOS transformation is directly reflected in the corresponding S-σ correlation. The shaded symbols indicate the experimental data presented in Figure 2.

Although the main purpose of this work is to highlight a common behavior across a wide range of materials, there are some insights to be obtained by looking closer into the different classes of polymers that were investigated. Based on our current data set of diverse polymers, we observe the following trends. In general, D-D (or D-π-D) polymers such as

P3HT and A-A (or A-π-A) polymers such as DPP-BTz exhibit high electrical conductivities but relatively small Seebeck coefficients (Table S2), leading to moderate power factors unless improved by postannealing or morphological optimization. In contrast, D−A polymers (e.g., PM6, PTB7-Th) tend to show higher Seebeck coefficients at comparable conductivities. This behavior is attributed to the acceptor unit (A) introducing additional energetic disorder along the polymer backbone, which enhances the Seebeck coefficient, while the donor unit (D) promotes efficient charge transfer with p-type dopants, maintaining reasonably high σ. A representative example is PDPP3T, which combines high conductivity with a large Seebeck coefficient. Our measurements show that its localization length is significantly larger than that of PM6, suggesting that polymers with enhanced charge delocalization and moderate energetic disorder are better suited to balance σ and S, thereby achieving higher power factors.

The discussion can be translated into a design rule for reaching high performance in OTE materials. Since PF $\propto \sigma^{1/2}$ until the roll-off point at max PF, one should aim to shift the roll-off to higher doping levels. This can be accomplished by suppressing carrier−carrier interactions to reduce the impact of the soft Coulomb gap on thermoelectric performance. Since the Coulomb interaction scales as $E_C = e/4\pi\varepsilon_r\varepsilon_0 a$, two strategies to achieve this can be identified, as depicted in Figure 4g,h (and Figure S22 in the SI). First, by increasing the relative permittivity (ε_r) of the material, for example by introducing polarizable side chains, and, second, by promoting charge delocalization, e.g., by planarizing the backbone, to increase the effective separation between carriers.

Moreover, the emergence of a soft Coulomb gap at high doping levels is a general phenomenon in disordered organic semiconductors. Thus, the insights gained here are not limited to organic thermoelectrics but are broadly relevant to other doping-driven applications, including organic light-emitting diodes (OLEDs), organic field-effect transistors (OFETs), and other emerging devices.

In summary, we have conducted a comprehensive study by combining experimental measurements with kinetic Monte Carlo simulations to elucidate the fundamental mechanisms governing the correlation between the Seebeck coefficient and electrical conductivity in doped conjugated polymers. Our findings reveal that the observed transition from the empirical power law $S \propto \sigma^{-0.25}$ to the logarithmic dependence $S \propto \ln(\sigma)$ originates from the formation of a soft Coulomb gap in the renormalized Gaussian density of states (DOS). This transition reflects the dominance of carrier−carrier Coulomb interactions. Furthermore, we demonstrate that although this transition is universally observed across various material systems, its exact shape and the position of the S-σ roll-off are critically dependent on the material's dielectric properties and degree of energetic disorder. These findings offer critical mechanistic insights into the interplay between electronic structure, charge transport, and thermoelectric performance, establishing a solid foundation for the rational design of next-generation high-performance organic thermoelectric materials. Specifically, the ability to shift the roll-off to higher doping concentrations is identified as a key factor for achieving enhanced power factors. Based on these insights, we propose a design strategy for developing high-performance OTE materials minimizing Coulomb interactions by either increasing the relative permittivity or by promoting charge delocalization.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsenergylett.5c03040.

Experimental section (including materials, device fabrication, conductivity and Seebeck coefficient measurement); Temperature dependence of conductivity and model fittings; kMC simulation results for temperature-dependent conductivities, DOS changes, and thermoelectric characteristics (PDF)

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Notes

The authors declare no competing financial interest.

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